

Exact $4n/3$ Compress-with-Another Threshold for an Explicit Completion of a $3k + 6$ -State Binary Family

Ryutaro Yonezu
Independent Researcher

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Abstract

Szykuła’s 2026 open-problems survey presents a binary family $\mathcal{A}_k$ on $n = 3k + 6$ states and conjectures that the shortest word compressing q_0 with another state has length $4n/3$. The displayed transition definition, however, does not specify all transitions required for a complete deterministic semiautomaton, and the displayed candidate word has length $4k + 9$ although $4n/3 = 4k + 8$. We therefore do not claim to resolve that conjecture as stated. Instead, we define an explicit total completion $\hat{\mathcal{A}}_k$ that agrees with every displayed transition and specifies the missing cases separately. For this explicitly defined family we prove that the compress-with-another threshold of q_0 is exactly

$$\text{cat}_{\hat{\mathcal{A}}_k}(q_0) = 4k + 8 = \frac{4n}{3}.$$

An explicit shortest witness is

$$c_k = (ba)^{k+2}(ab)^{k+2}.$$

The lower bound is obtained by an exact reverse-layer description in the unordered pair automaton. We also give an explicit reset word, proving that every $\hat{\mathcal{A}}_k$ is synchronizing. The general lower-bound proof is analytic for $k \geq 5$; the four small cases $k = 1, 2, 3, 4$ are checked by an exact finite verifier included with the manuscript. No claim is made that the completion used here is identical to the completion intended in the original source.

1 Introduction

For a deterministic finite complete semiautomaton $\mathcal{A} = (Q, \Sigma, \delta)$ and a state $q \in Q$, define the *compress-with-another threshold* of q as the minimum length of a word w such that

$$\delta(q, w) = \delta(p, w)$$

for some $p \neq q$. Szykuła recently isolated this parameter as a natural synchronization primitive and conjectured a linear upper bound for synchronizing strongly connected automata [1]. The same survey presents a particular binary family on $3k + 6$ states for which the threshold of q_0 is conjectured to equal $4n/3$ [1, Sec. 2.3, Conj. 8]; the construction is attributed there to Dzyga’s 2018 master’s thesis [2].

There is a source-level obstacle to treating the displayed construction as a fully specified object. The survey defines its ambient model to be complete deterministic semiautomata [1, Sec. 1.1], but the displayed piecewise transition formula for $\mathcal{A}_k$ leaves $\delta(q_0, b)$ unspecified and, depending on the parity of the terminal state q_{3k+5} , leaves one terminal transition unspecified as well. Independently, Conjecture 8 prints the word

$$(ab)^{k+2}a(ab)^{k+2}$$

and labels its length as $4n/3$. The printed word has length $4k + 9$, whereas $n = 3k + 6$ gives $4n/3 = 4k + 8$.

The purpose of this note is therefore deliberately narrower than “resolving Conjecture 8”. We state a total automaton explicitly, agreeing with all displayed transition cases and adding only the missing transitions. For this total automaton we prove the exact $4n/3$ threshold and synchronizability. This separates a mathematical statement that can be checked from a provenance question that cannot be settled from the displayed formula alone.

Main result. Let $\hat{\mathcal{A}}_k$ be the explicit completion in Definition 1. Then for every $k \geq 1$, $\hat{\mathcal{A}}_k$ is strongly connected and synchronizing, and

$$\text{cat}_{\hat{\mathcal{A}}_k}(q_0) = 4k + 8 = \frac{4}{3}(3k + 6).$$

The word $c_k = (ba)^{k+2}(ab)^{k+2}$ attains the bound. For $k \geq 5$ the lower bound is proved by an exact symbolic reverse-pair analysis; the cases $k \leq 4$ are finite and are independently checked by the accompanying verifier.

2 The explicit completion

Fix $k \geq 1$ and put

$$n = 3k + 6, \quad A = 2k + 4, \quad B = 2k + 5, \quad M = 3k + 5 = n - 1.$$

Thus A is even and B is odd.

Definition 1 (Explicit completion). The automaton $\hat{\mathcal{A}}_k = (Q, \{a, b\}, \hat{\delta}_k)$ has

$$Q = \{q_0, q_1, \dots, q_M\}.$$

Its action under a is

$$q_i a = \begin{cases} q_0, & i = 0, \\ q_{i+1}, & 1 \leq i \leq 2k + 3, \text{ } i \text{ odd}, \\ q_{i-1}, & 1 \leq i \leq 2k + 3, \text{ } i \text{ even}, \\ q_{i+1}, & A \leq i < M, \text{ } i \text{ even}, \\ q_{i-1}, & 2k + 7 \leq i < M, \text{ } i \text{ odd}, \\ q_3, & i = B, \\ q_M, & i = M, \text{ } M \text{ even}, \\ q_{M-1}, & i = M, \text{ } M \text{ odd}, \end{cases}$$

and its action under b is

$$q_i b = \begin{cases} q_1, & i = 0, \\ q_{i+1}, & 1 \leq i \leq 2k + 3, \text{ } i \text{ even}, \\ q_{i-1}, & 1 \leq i \leq 2k + 3, \text{ } i \text{ odd}, \\ q_{i+1}, & B \leq i < M, \text{ } i \text{ odd}, \\ q_{i-1}, & 2k + 6 \leq i < M, \text{ } i \text{ even}, \\ q_2, & i = A, \\ q_M, & i = M, \text{ } M \text{ odd}, \\ q_{M-1}, & i = M, \text{ } M \text{ even}. \end{cases}$$

Every case printed in the transition display of [1, Sec. 2.3] is retained. The additional assignments are $q_0b = q_1$ and the parity-complementary terminal transition $q_M \mapsto q_{M-1}$ for whichever letter does not already have the printed terminal self-loop. These additional assignments are part of the definition of $\hat{\mathcal{A}}_k$; we do not assert that they were the intended missing assignments in [1, 2].

Proposition 1. *Each letter of $\hat{\mathcal{A}}_k$ has rank $n - 1$. The only two-element fibers are*

$$q_4a = q_Ba = q_3, \quad q_3b = q_Ab = q_2.$$

Consequently a non-diagonal pair can become diagonal in one step only from

$$C_a = \{q_4, q_B\} \quad \text{under } a, \quad C_b = \{q_3, q_A\} \quad \text{under } b.$$

Proof. Inspection of Definition 1 shows that all transitions except the displayed portal pairs are injective on their local adjacent-pair blocks. Under a , the two predecessors of q_3 are q_4 and q_B , and the unique missing image is q_{2k+3} . Under b , the two predecessors of q_2 are q_3 and q_A , and the unique missing image is q_A . No other image has two predecessors. The final statement follows immediately. $\square$

Proposition 2. *For every $k \geq 1$, $\hat{\mathcal{A}}_k$ is strongly connected.*

Proof. Starting at q_0 , one reaches q_1 by b , then q_2 by a , then q_3 by b , and so on along the main arm by alternating a and b until q_{2k+3} . The letter a then reaches q_A and a second a reaches q_B . From q_B , the alternating forward transitions along the tail reach every state up to q_M . Thus every state is reachable from q_0 .

Conversely, every main-arm state can be moved downward to q_1 by the opposite adjacent-pair transitions and then to q_0 by b . Every tail state can be moved downward along the tail; $q_Ba = q_3$ enters the main arm. At q_M , the non-looping letter sends q_M to q_{M-1} , after which the same downward argument applies. Hence every state reaches q_0 . $\square$

3 The $4n/3$ upper bound

Define

$$c_k = (ba)^{k+2}(ab)^{k+2}.$$

Its length is $4k + 8 = 4n/3$.

Lemma 1. *For every $k \geq 1$,*

$$q_0c_k = q_2c_k = q_Bc_k = q_2.$$

In particular c_k compresses q_0 with another state.

Proof. It is enough to follow the pair $\{q_0, q_2\}$. During the first $2k + 4$ letters, namely $(ba)^{k+2}$, the pair evolves as

$$\{q_0, q_2\}, \{q_1, q_3\}, \{q_2, q_4\}, \dots, \{q_{2k+2}, q_A\}, \{q_2, q_{2k+3}\}, \{q_1, q_A\}.$$

Now apply the second block $(ab)^{k+2}$. For the first $k + 1$ letters of this block the pair is

$$\{q_{s+1}, q_{B+s-1}\} \quad (1 \leq s \leq k + 1),$$

so it reaches $\{q_{k+2}, q_M\}$. One more letter gives $\{q_{k+3}, q_M\}$, after which the second coordinate moves downward while the first moves upward. More precisely, after s letters of the second block, for $k + 2 \leq s \leq 2k + 2$, the pair is

$$\{q_{s+1}, q_{M-s+k+2}\}.$$

The next letter gives $C_b = \{q_3, q_A\}$, and the final b maps both states to q_2 . Thus $q_0c_k = q_2c_k = q_2$. A direct substitution from q_B through the same word gives $q_Bc_k = q_2$ as well. $\square$

Remark 1 (The printed candidate word). The candidate word printed in Conjecture 8 of [1] is

$$(ab)^{k+2}a(ab)^{k+2}.$$

Using $a(ba)^m = (ab)^ma$, this word is exactly ac_k . Thus the printed word is one letter longer than c_k , matching the one-letter discrepancy between its literal length $4k + 9$ and the annotated value $4n/3 = 4k + 8$.

4 Reverse pair layers and the lower bound

Let $\mathcal{P}$ be the unordered non-diagonal pair automaton of $\widehat{\mathcal{A}}_k$. For a pair $P \in \mathcal{P}$ and a letter $x \in \{a, b\}$, write Px for the unordered image pair when the image is non-diagonal. Let

$$\mathcal{C} = \{C_a, C_b\}.$$

For $t \geq 0$, let $\mathcal{L}_t$ be the set of non-diagonal pairs whose shortest path in $\mathcal{P}$ to $\mathcal{C}$ has length exactly t .

By Proposition 1, if a pair containing q_0 is compressed by a word of length ℓ , then after its first $\ell - 1$ letters it must lie in $\mathcal{C}$. Hence a lower bound on the distance from every $\{q_0, q_j\}$ to $\mathcal{C}$ gives a lower bound on the compress-with-another threshold of q_0 .

The next lemma gives the exact reverse frontier until a pair containing q_0 first appears. We identify $\{q_i, q_j\}$ with the index pair $\{i, j\}$ to shorten notation.

Lemma 2 (Exact reverse layers). *Assume $k \geq 5$. With $A = 2k + 4$, $B = A + 1$, and $M = 3k + 5$, the layers $\mathcal{L}_t$ for $0 \leq t \leq 4k + 6$ are exactly the following sets.*

Phase I: $0 \leq t \leq k$.

$$\mathcal{L}_t = \{\{3 + t, A - t\}, \{4 + t, B + t\}\} \cup \begin{cases} \emptyset, & t = 0, \\ \{\{A - t, A + t\}\}, & 1 \leq t \leq k. \end{cases}$$

Phase II: $t = k + s$, $1 \leq s \leq k + 1$. Let

$$F_1 = \{k + 3, M\}, \quad F_s = \{k + 4 - s, M - s + 2\} \quad (s \geq 2),$$

and, for $1 \leq s \leq k - 1$,

$$G_s = \{k + 4 + s, M - s + 1\}.$$

Then

$$\mathcal{L}_{k+s} = \{F_s\} \cup \begin{cases} \{G_s\}, & 1 \leq s \leq k - 1, \\ \emptyset, & k \leq s \leq k + 1. \end{cases}$$

Phase III: $t = 2k + 2 + r$, $0 \leq r \leq k + 2$. Set

$$H_0 = \{2, B\}, \quad H_r = \{r, A + 1 - r\} \quad (r \geq 1).$$

Include additionally

$$J_1 = \{A, A + 2\}, \quad J_r = \{A + 1 - r, A + 3 - r\} \quad (r \geq 3),$$

(no J_0 or J_2),

$$K_r = \{A + 1 - r, A + r - 3\} \quad (r \geq 4),$$

and

$$L_r = \{A + 1 + r, A + 3 + r\} \quad (0 \leq r \leq k - 2).$$

At $r = k - 1$ include also $\{M - 1, M\}$. Then $\mathcal{L}_{2k+2+r}$ is exactly the union of the terms whose stated ranges contain r .

Phase IV: $t = 3k + 5 + s$, $0 \leq s \leq k + 1$. Let

$$U_s = \{k + 2 - s, k + 4 - s\},$$

and

$$V_0 = \{k + 2, M - 1\}, \quad V_1 = \{k + 1, M\}, \quad V_s = \{k + 2 - s, M - s + 2\} \quad (s \geq 2).$$

Then $\mathcal{L}_{3k+5+s}$ contains U_s and V_s . At $s = k$ it contains in addition

$$\{6, B\}, \quad \{B, B + 4\},$$

and at $s = k + 1$ it contains in addition

$$\{5, A\}, \quad \{7, A + 2\}, \quad \{A, A + 4\}, \quad \{A + 2, A + 6\}.$$

There are no other pairs in these layers.

Finally, at distance $4k + 7$ the pairs $\{0, 2\}$ and $\{0, B\}$ occur.

Proof. Let $R_t = \bigcup_{j=0}^t \mathcal{L}_j$ and let $\text{Pred}(X)$ denote the set of non-diagonal pairs that map by one letter into a pair in X . Since the pair automaton is deterministic forward, the exact reverse breadth-first recurrence is

$$\mathcal{L}_{t+1} = \text{Pred}(\mathcal{L}_t) \setminus R_t. \quad (1)$$

Starting from

$$\mathcal{L}_0 = \{\{3, A\}, \{4, B\}\},$$

substitution of the transition rules in Definition 1 into (1) gives the four displayed phases. Away from the portal indices $3, 4, A, B$ and the endpoint M , each inverse step simply moves one coordinate by one position along an adjacent-pair block; this yields the affine branches in Phases I–III. The exceptional predecessor events are exactly: the disappearance of the J branch at $r = 2$ because that pair already lies in an earlier layer; creation of $\{M - 1, M\}$ at $r = k - 1$; and the two boundary bursts at $s = k$ and $s = k + 1$ in Phase IV. Substituting these boundary indices directly into Definition 1 gives precisely the extra pairs listed above and no others.

For completeness, the recurrence can be checked phase by phase without inference: applying both inverse letter maps to every pair in the displayed $\mathcal{L}_t$, deleting pairs already in R_t , produces exactly the displayed $\mathcal{L}_{t+1}$. All expressions stay in the indicated index ranges for $k \geq 5$. At the next step, $s = k + 2$, the two generic Phase-IV branches give

$$U_{k+2} = \{0, 2\}, \quad V_{k+2} = \{0, B\}.$$

Thus a pair containing state 0 first appears at distance $4k + 7$. An independent exact verifier implementing recurrence (1) is included with the manuscript. $\square$

Corollary 1. For $k \geq 5$, every word compressing q_0 with another state has length at least $4k + 8$.

Proof. Every pair in the layers $\mathcal{L}_0, \dots, \mathcal{L}_{4k+6}$ has both indices positive by Lemma 2. Hence no pair $\{q_0, q_j\}$ reaches $\mathcal{C}$ in fewer than $4k + 7$ letters. By Proposition 1, one further letter is necessary to make the pair diagonal. Therefore at least $4k + 8$ letters are required. $\square$

Theorem 1 (Exact threshold). For every $k \geq 5$,

$$\text{cat}_{\widehat{\mathcal{A}}_k}(q_0) = 4k + 8 = \frac{4n}{3}.$$

The word $c_k = (ba)^{k+2}(ab)^{k+2}$ is shortest.

Proof. Lemma 1 gives the upper bound $4k + 8$ and Corollary 1 gives the matching lower bound. $\square$

5 Synchronizability

We next give an explicit reset word. For odd $r \geq 1$, define

$$B_r = (ba)^{(r-1)/2}b.$$

Set

$$R_k = B_{2k+3}a^2B_{2k+1}a^4B_{2k-1} \cdot \prod_{j=0}^k (a^2B_{4k+3-2j}) \cdot a^2B_{2k+3}, \quad (2)$$

where the product is read from left to right in increasing j .

Lemma 3. *For every $k \geq 3$,*

$$QR_k = \{q_2\}.$$

Hence $\widehat{\mathcal{A}}_k$ is synchronizing. The same statement for $k = 1, 2$ follows by direct substitution in (2).

Proof. Write $O[u, v] = \{q_i : u \leq i \leq v, i \text{ odd}\}$. Direct application of the adjacent-pair transitions gives

$$QB_{2k+3} = \{q_0, q_1, q_2, q_3\} \cup O[5, 2k+3],$$

then

$$QB_{2k+3}a^2 = \{q_0, q_1, q_2, q_3, q_B\} \cup O[5, 2k+1],$$

and

$$QB_{2k+3}a^2B_{2k+1} = \{q_0, q_1, q_3\} \cup O[5, B].$$

After the next two blocks a^4B_{2k-1} , the image is

$$S_{-1} = \{q_0, q_1, q_2, q_3, q_{2k+3}\} \cup O[5, 2k-1].$$

For $0 \leq j \leq k-3$, the block $a^2B_{4k+3-2j}$ deletes the current largest element of the odd interval and leaves all other displayed states in the same canonical positions, so after block j the image is

$$\{q_0, q_1, q_2, q_3, q_{2k+3}\} \cup O[5, 2k-3-2j].$$

At this point, and after the remaining three blocks in the product of (2), the successive images are

$$\{q_0, q_1, q_2, q_3, q_{2k+3}\}, \quad \{q_0, q_1, q_2, q_{2k+3}\}, \quad \{q_0, q_2, q_{2k+3}\}, \quad \{q_2, q_{2k+3}\}.$$

The final block a^2B_{2k+3} maps both remaining states to q_2 . Every displayed identity follows by substituting the corresponding indices into Definition 1. Thus $QR_k = \{q_2\}$. $\square$

The length of (2) is

$$|R_k| = 3k^2 + 16k + 19.$$

We make no claim here that this reset word is shortest.

6 The four small cases and computational audit

The symbolic layer description in Lemma 2 was stated for $k \geq 5$ to avoid degeneracies where branches collide or empty index ranges overlap. The exact finite pair automata for $k = 1, 2, 3, 4$ give the following minimum distances from a pair containing q_0 to the diagonal:

k	n	$\min_{j \neq 0} d(\{0, j\}, \text{diagonal})$	minimizing j
1	9	12	2, 7
2	12	16	2, 9
3	15	20	2, 11
4	18	24	2, 13

Thus the same value $4k + 8$ holds in all four cases. The accompanying script constructs the automaton from Definition 1, performs exact multi-source breadth-first search in the unordered pair graph, checks the reverse-layer formulas for arbitrary supplied $k \geq 5$, and verifies the reset word. A fresh manuscript-side audit run checked all items for $1 \leq k \leq 100$ (including the symbolic-layer template for $5 \leq k \leq 100$) and returned **PASS**; this finite sweep is diagnostic and is not used as the general proof.

Combining the finite cases with Theorem 1 gives the following statement for the explicit completion.

Corollary 2. *For every $k \geq 1$, the explicitly defined automaton $\hat{\mathcal{A}}_k$ is strongly connected and synchronizing, and*

$$\text{cat}_{\hat{\mathcal{A}}_k}(q_0) = 4k + 8 = \frac{4n}{3}.$$

For $k \geq 5$ the threshold equality is proved analytically above; for $1 \leq k \leq 4$ it includes the exact finite verification reported in this section.

7 Claim boundary and relation to Conjecture 8

The mathematical result in this paper concerns Definition 1, not an inferred unpublished transition table. Three source facts motivate this separation:

1. Szykuła’s paper works throughout with complete deterministic semiautomata [1, Sec. 1.1].
2. The displayed definition of $\mathcal{A}_k$ in Sec. 2.3 does not assign q_0b , and it gives only one parity case for the terminal state under each letter; therefore one terminal action is unassigned for every k [1, Sec. 2.3].
3. The candidate word printed in Conjecture 8 has literal length $4k + 9$, while the same sentence gives the claimed length $4n/3 = 4k + 8$ [1, Conj. 8].

The family is attributed in that paper to Dzyga’s 2018 master’s thesis [2]. This manuscript does not use the thesis to identify the missing transitions and does not claim that $\hat{\mathcal{A}}_k$ is the author’s intended completion. Accordingly, the result should not be cited as a proof of Conjecture 8 without an independent source-level identification of the intended total transition function.

If such an identification is later established and it matches Definition 1, then Corollary 2 would immediately supply the corresponding mathematical part of Conjecture 8, with the one-letter-shorter word c_k in place of the literally printed word.

8 Conclusion

We isolated the provenance issue in a recently proposed extremal family from the mathematical question that can be posed on an explicit total automaton. For the completion $\hat{\mathcal{A}}_k$ stated here, the state q_0 has exact compress-with-another threshold $4n/3$, witnessed by $(ba)^{k+2}(ab)^{k+2}$, and the family is synchronizing by an explicit reset construction. The lower bound follows from an exact reverse-pair frontier whose first encounter with a pair containing q_0 occurs at distance $4k + 7$ from the two final-collision pairs.

The remaining issue is not a missing mathematical inequality but source identification: whether the total transition function intended in the original construction is the completion studied here. Until that is established, the conservative statement is the one proved in this manuscript.

Reproducibility note

The supplementary code is split by purpose and uses only the Python 3 standard library. `completion_model.py` contains the explicit total transition system and shared exact routines. `experiment_exact_threshold.py` performs the main finite experiment: it computes the exact q_0 compress-with-another threshold by pair BFS, verifies the length- $4k + 8$ witness and the reset construction, and writes a CSV audit table. `verify_reverse_layers.py` independently compares the exact reverse-pair BFS frontier with the closed-form layer template used in Lemma 2. A fresh sweep through $k = 100$ passes both programs. These computations are audit aids; the $k \geq 5$ lower bound in the manuscript is the symbolic argument of Lemma 2, not an inference from a finite sweep.

References

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